

Programming part

1-In the following you will find the description of a booking system for train journeys (we will refer to train journeys as trains for brevity in the rest of description). All fields must be private, and all methods listed for each question must be public. Any auxiliary helper method you create must be private. You may assume that all arguments used in method calls have valid values, so you do not need to use defensive programming (check for validity of arguments) (65 points)

- (1.1) Define a class called **Train**. In the class add the fields: **destination** of type **String** representing the destination of the train, **capacity** of type **int** representing the maximum capacity of the train (shows maximum number of tickets that can be booked for the train in total), **totalBooked** of type **int** representing the current total number of booked tickets for the train, and **ticketPrice** of type **int** representing the price of the ticket for the train.
- (1.2) Implement the constructor **Train(String destination, int capacity, int ticketPrice)** such that it initializes the fields **destination**, **capacity**, and **ticketPrice** with the value of the corresponding parameters of the constructor. Furthermore, it initializes **totalBooked** to 0.
- (1.3) In class **Train**, implement the methods **getDestination()** and **getTicketPrice()** which return the value of the fields **destination** and **ticketPrice**, respectively.
- (1.4) In class **Train**, implement the method **hasCapacity(int num)** that returns **true**, if there are **num** tickets available for the train (considering the maximum capacity and current total booked tickets). Otherwise, it returns **false**.
- (1.5) In class **Train**, implement a method **book(int num)**, that books **num** tickets, increasing **totalBooked** accordingly, if the train has enough capacity and returns **true**. Otherwise, it returns **false**. For full credit you must use the **hasCapacity** method.
- (1.6) In class **Train**, implement the method **print()** that prints the information of the train as: "**Destination: <destination>, capacity: <capacity>, total booked tickets: <totalBooked>, ticket price: <ticketPrice>**", where **<x>** represents the value of the corresponding field **x**.
- (1.7) Implement a class **BookingSystem**. In the class add a field **trains** of type **List<Train>**. Implement the constructor of the class **BookingSystem()**, which initializes the **trains** list with an empty list.

- (1.8) In class `BookingSystem`, implement a method `addTrain(Train train)` that adds `train` to `trains` list.
- (1.9) In class `BookingSystem`, implement a method `bookTrain(String dest, int num)` that finds the first train to destination `dest` in the `trains` list that has enough capacity for booking `num` tickets. If such train is found the tickets are booked in the train and the method prints **"Your tickets are booked"**. Otherwise, it prints **"Booking not possible"**.
- (1.10) In class `BookingSystem`, implement a method `removeFull()` that removes trains which are fully booked (have no more capacity) from the `trains` list.
- (1.11) In class `BookingSystem`, implement a method `showTrains(int maxPrice)` that prints per line the information of all trains in `trains` list that have ticket price less than `maxPrice`. The information of each train should be printed with the format described in task (1.6).

Add the `Train.java` and `BookingSystem.java` files in one folder called `YourName_YourStudentNumber`. Upload the compressed folder as a `.zip` file via the link provided for submitting the midterm answers.

Multiple Choice part

2-Consider the following class:

```
public class Group{
    private int trace;
    private Teacher member;
    private Book book;

    public void add(int c){...}
    public int getC(){...}
}
```

Select the correct statement(s). (3 points)

- (a) The class has three fields named `trace`, `Teacher`, and `Book`
- (b) The `add` method can return either a `String` or an `int` value
- (c) The `getC` method has no parameters
- (d) The objects of the class can be created using the statement `new Group()`

3-Consider the code below.

```
public class Machine{
    private int count;

    (*)
}
```

Select option(s) that can replace (*) in the code above such that: `new Machine(5)` creates an object of type `Machine` in which the value of the field `count` is initially 5. (2 points)

- (a) `public Machine(int c){ int count=5; }`
- (b) `public Machine(int c){ int count=c; }`
- (c) `public Machine(int count){ this.count=count; }`
- (d) `public Machine(int count){ count=5; }`

4-Assume that `nums` is an array of `int` in which all elements have value zero. Which of the following option(s) increase the value of the element at index `i` in the array by one?

(2 points)

- (a) `++nums[i]`
- (b) `i++`
- (c) `nums[i]=nums[i]+1`
- (d) `nums[i]=nums[i+1]`

5-Assume that `x`, `y`, and `z` are of type `int`. Which of the following option(s) assigns the same value to variable `x` as the expression below (for all possible values of these variables)?

(2 points)

`x = y+1>0 ? y*z : y+z;`

- (a) `if(x==y+1) {x=y;}
else {x=y+z;}`
- (b) `if(x==y) {x=y+z;}
else{x=y;}`
- (c) `if(y+1>0) {x=y+z;}
else{x=y*z;}`
- (d) `if(y+1>0) {x=y*z;}
else{x=y+z;}`

6-Consider the code below:

```
public class C{  
    public static boolean xor(boolean a,boolean b){...}  
}
```

Assume that the `xor` method returns `false` if its parameters `a` and `b` have the same value and `true` otherwise. Select the option(s) that is/are suitable for testing this method.

(2 points)

- (a) `assertEquals(true,C.xor(true,false))`
- (b) `assertEquals(false,C.xor(false,false))`
- (c) `assertEquals(false,C.xor(false,true))`
- (d) `assertEquals(true,C.xor(true,true))`

7-Assume that `age` is a variable of type `int`. Select the option(s) that increase the value of `age` by 6. (2 points)

- (a) `for(int i=1; i<=5; i++){age=age+1;}`
- (b) `for(int i=0; i<=5; i++){age++;}`
- (c) `for(int i=10; i>5;i--){++age;}`
- (d) `for(int i=6; i>=1; i--){age=age+1;}`

8-Consider the code below. What is the length of the array `arr`? (2 points)

```
String[] arr = new String[3];  
arr[0]="2";  
arr[1]=arr[0];
```

- (a) 1
- (b) 0
- (c) 3
- (d) 2

9-Assume that `x` is a variable of type `byte`. Which of the following option(s) is/are a valid value for `x`? (2 points)

- (a) 128
- (b) 100
- (c) -129
- (d) -1

10-Which of the following option(s) define a variable with a reference type? (3 points)

- (a) `ArrayList<Integer> x;`
- (b) `int x;`
- (c) `int[] x;`
- (d) `char x;`

11- Which of the following statement(s) is/are true? (2 points)

- (a) Every class must define at least one field
- (b) Low cohesion is desirable in good class design
- (c) Code duplication is not desirable in good class design
- (d) A class can have several constructors

12- Consider the code below.

```
public class Item{
    public String print(){return "This is an item";}
}
public class SpecItem extends Item{
    public String print(){return "0"+super.print();}
}
public class BulkItem extends SpecItem{
    public String print(){return super.print()+"1";}
}
```

Which `print` method(s) is/are executed as a result of successfully running the code below?
(2 points)

```
Item a;
var b = new BulkItem();
a=b;
a.print();
```

- (a) Only the `print` method in class `BulkItem`
- (b) The `print` methods in classes `BulkItem`, `SpecItem` and `Item`
- (c) Only the `print` method in class `Item`
- (d) Only the `print` methods in classes `BulkItem` and `SpecItem`

13- Consider the code below.

```
public interface Card{
    public int balance();
}
public interface MasterCard extends Card{
    public String getNumber();
    public void transfer(int m);
}
public class MyCard implements MasterCard{...}
```

Select the correct option(s). (2 points)

- (a) The class `MyCard` is an abstract class
- (b) The class `MyCard` has at most three methods
- (c) The class `MyCard` has at least three methods
- (d) The class `MyCard` cannot have a method with signature `int getNumber()`

14- Consider the following class definition.

```
public class Cycle{
    private static int tick;

    public Cycle(){tick=tick+5;}
    public static void init(){tick=0;}
    public void increase(){tick++;}
    public int getTick(){return tick;}
}
```

What value is returned by the method call in the last line in the code below? (2 points)

```
Cycle.init();
var a = new Cycle();
var b = new Cycle();
b.increase();
a.getTick();
```

- (a) 5
- (b) 11
- (c) 10
- (d) 6

15- Assume that `myMap` is a variable of type `TreeMap<Integer, Integer>`. Given that `myMap` is empty before running the code below; select the option(s) that correctly represent the mappings (from keys to values) in `myMap` after running the code. (3 points)

```
myMap.put(16,1);
myMap.put(10,5);
myMap.put(16,myMap.get(16)+1);
myMap.put(2,5);
myMap.put(1,11);
myMap.put(10,3);
myMap.remove(2);
```

- (a) $16 \rightarrow 2$
- (b) $2 \rightarrow 5$
- (c) $1 \rightarrow 11$
- (d) $1 \rightarrow 3$

16- Consider the code below.

```
public class Book{...}
public class OnlineBook extends Book{...}
public class Main{
    public static void myMethod(List<Integer> x, Book y){...}
}
```

Given that the method call `Main.myMethod(a, b)` runs successfully; which of the following option(s) show valid types for `a` and `b`, respectively? (2 points)

- (a) `OnlineBook, Integer`
- (b) `ArrayList<Integer>, Book`
- (c) `Integer, Book`
- (d) `ArrayList<Integer>, OnlineBook`

17- Consider the following class definition.

```
public class Person{
    private String name;
    public Person(String n){name=n;}
    public void changeName(String n){name=n;}
    public String getName(){return name;}
}
```

Select statement(s) that is/are correct about the code below. (2 points)

```
Person p0=new Person("Alice");
Person p1=p0;
final Person[] arr={p0,p1};
arr[1].changeName("Kathrin");
```

- (a) `arr[0].equals(arr[1])` is **true** after running the code
- (b) `arr[0].equals("Alice")` is **true** after running the code
- (c) The code does not compile successfully as the content of a **final** array cannot change
- (d) `p0.getName()` will return `Kathrin` after running the code